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Critical Overflow in Weakly Coupled Two-Choice Allocation

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Abstract

Community totals miss a leading contribution to overflow in two-choice allocation with finite bucket capacity. For two weakly coupled communities of q buckets each, we identify this positive order-q contribution through the internal load distribution. The omitted overflow equals the sum of the smaller of excess load and unused capacity within each community. At nominal total capacity, expected overflow grows linearly in q, although its fraction of arrivals tends to zero. Each arrival joins the less loaded of two candidates, with ties favoring the left side. Internal candidates are sampled uniformly from a complete bipartite graph with two equal sides. Cross-community sampling has probability ε and uses predetermined kernels with uniform endpoint marginals. Starting from empty buckets, we let $q \to \infty$ with $\varepsilon q \to \lambda < \infty$ and integer capacity c_q satisfying $2c_q/q \to \kappa > 0$. We prove a joint limit for community imbalance and overflow on the q^2 arrival scale, including leading mean asymptotics. A scalar diffusion captures imbalance, while an attracting periodic load profile determines overflow. The diffusion averages the lattice phase, while overflow retains its observation phase. At nominal fill and zero normalized imbalance, the profile readout is approximately 0.6313. Sensitivity experiments find positive mean omitted overflow under left and fair ties across four internal candidate graphs. At $q = 512$ and $\kappa = \lambda = 1$, the left-tie mean omitted overflow is 5.8 times the complete-bipartite mean for cycles and 21.4 times for disconnected matchings. Across $q = 128, 256, 512$, mean omitted overflow divided by q increases for both topologies.

Keywords

Randomized allocation; Two choices; Load balancing; Capacity overflow; Diffusion approximation; Internal load distribution

1 Introduction

Two-choice allocation assigns each item to the less loaded of two candidates [1]. With finite bucket capacity, the performance question is how many items overflow as the system fills. For two communities of q buckets each, we prove that community imbalance alone misses a positive order-q contribution in the critical limit. The load distribution within each community determines this missing term. We derive a joint critical law that retains both sources of overflow.

Consider two communities of $q \geq 2$ buckets with integer capacity $c \geq 1$. In the first, two buckets have loads $c-1$ and $c+1$; all remaining buckets have load c. In the second, every bucket has load c. Both totals are qc, but the first community has one unit of overflow and the second has none. For bucket capacity c, let Z_c denote total overflow and M_s the total load in community C_s. Let E_s and U_s denote its total load above capacity and total unused capacity. The omitted overflow is

$$\begin{aligned} G_c &:= Z_c - \sum_s (M_s - |C_s|c)_+ \\ &= \sum_s \min\{E_s, U_s\}. \end{aligned} \tag{1.1}$$

The nonnegative gap G_c measures the excess load hidden by unused capacity within the same community. It is positive exactly when some community contains buckets both above and below capacity. Our critical law shows that this local imbalance contributes at leading order. Section 7.2 gives the equivalent absolute-deviation form.

Buckets represent finite-capacity allocation locations, grouped into two communities. Candidate edges specify which pairs an arrival can compare. Each community samples internal candidates uniformly from a complete bipartite graph. A fraction ε of arrivals sample across the communities through any predetermined doubly stochastic kernel. Both fixed matchings and product sampling have the required uniform endpoint marginals. Ties favor the left endpoint, as in the internal two-group Always-Go-Left rule of Vöcking [2]. Internal sampling determines how new load spreads within a community, while cross sampling controls interaction between communities.

We write c_q for bucket capacity when it depends on q. With empty starts and integer capacity $c_q \asymp q$, critical fill occurs at $n_* = 2q\,c_q \asymp q^2$ arrivals. Without cross sampling, community choices produce order-q imbalance fluctuations. Cross arrivals can contribute order εq to the cumulative normalized drift. The fluctuation and drift scales motivate the regime $q \to \infty$, $\varepsilon q \to \lambda < \infty$, and $2c_q/q \to \kappa > 0$.

Our critical law combines a scalar diffusion for community imbalance with an attracting periodic profile for internal loads. At critical fill, let d denote normalized imbalance. The limiting normalized overflow readout is $S_0(d)$, whereas normalized community excess is $|d|/2$. We prove $S_0(d) > |d|/2$ for every finite d. Corollary 2.4 converts this gap into a strictly positive leading correction to expected overflow.

At nominal total capacity, expected overflow grows linearly in q, while its fraction of arrivals tends to zero. A vanishing overflow frac-

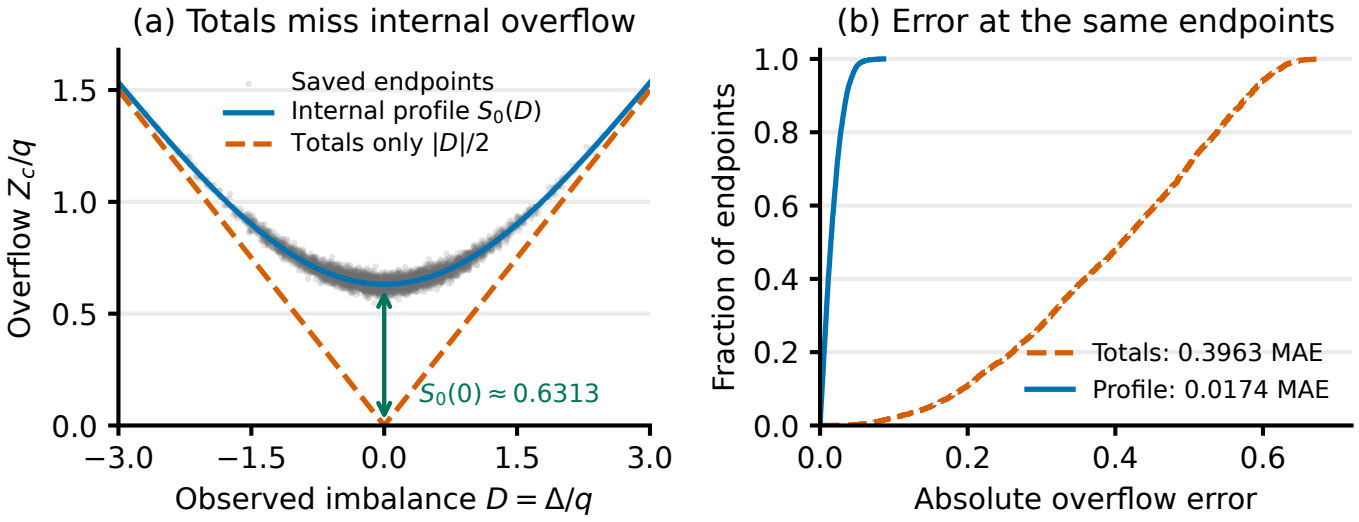


Fig. 1 Community totals miss internal overflow at critical fill. Left: 4096 saved endpoints at $q = 512$, $\kappa = 1$, $\lambda = 1$, with matching crosses. The solid curve is $S_0(D)$; the dashed curve is community excess $|D|/2$. The limiting readouts satisfy $S_0(d) > |d|/2$, separating internal overflow from community excess. Right: absolute-error distributions for the same endpoints, with mean absolute errors in the legend. Across these endpoints, the profile lowers mean absolute error by approximately 95.6% relative to community excess. The frozen profile lookup error is at most 0.000026.

tion therefore coexists with a growing absolute capacity cost.

Fig. 1 shows the missing internal contribution. Section 6 evaluates the profile readout and the terminal diffusion in separate comparisons.

1.1 Contributions

We establish a joint asymptotic law for capacity overflow in Theorems 2.1–2.3.

(i) **Diagnosing and quantifying overflow hidden by aggregation.** Equation (1.1) identifies the omitted overflow as the sum of the smaller of excess load and unused capacity within each community. Theorem 2.3 and Corollaries 2.4 and 2.5 quantify its positive order-q expected magnitude at nominal capacity. An attracting internal load profile, coupled to a scalar imbalance diffusion, determines the leading capacity cost.

(ii) **Robustness to cross-community connections.** For fixed cross connections with uniform endpoint marginals, the leading critical law agrees with product sampling (Theorem 2.1). The process coupling bounds both discrepancies by $o(q)$ in expected uniform distance over Tq^2 arrivals, simultaneously for all common capacities.

(iii) At $q = 512$ and $\kappa = \lambda = 1$, cycles and disconnected matchings have left-tie mean omitted overflow 5.8 and 21.4 times the complete-bipartite mean (Fig. 5). Across 24 tested configurations, mean omitted overflow is positive under left and fair ties for all four internal graphs. For cycles and disconnected matchings, mean omitted overflow divided by q increases across $q = 128, 256, 512$. A matching control gives order-$q^{3/2}$ expected overflow under weak cross sampling, showing that internal graph structure can also change the growth order (Supplement section SM9.2).

1.2 Relation to previous work

Existing rules and distribution equations. Vöcking [2] introduced Always-Go-Left. Berenbrink, Czumaj, Steger, and Vöcking [3] give sharp heavy-load deviation bounds for greedy and asymmetric allocation. Mitzenmacher [4] uses differential equations to describe full load distributions. We use this established distribution-level viewpoint. The additional requirements here are long-time attraction of a two-type integer-lattice profile and its coupling to a slow community imbalance.

Correlated choices and process comparison. Mitzenmacher [5, 6] shows asymptotic equivalence between double hashing and fully random balanced allocation, using fluid limits and a coupling through a biased intermediate process. The stated distribution results concern a constant number of choices and TN arrivals into N bins for fixed T. Our comparison covers Tq^2 arrivals with cross probability $O(q^{-1})$. The process bound then transfers the critical overflow law.

Graphs and queues. Peres, Talwar, and Wieder [7] study graphical allocation and limited choice. Bansal and Feldheim [8] give a strategy based on graph decompositions. Our rule compares only the two endpoint loads. We measure the overflow omitted by community totals. The sensitivity study connects this capacity readout to restricted internal graphs (Section 6.5). Queueing universality and diffusion limits [9–11] include service and use different scales. Here each item has two candidates and loads only increase.

Traveling profiles. Stolyar [12, 13] develops traveling-wave and stationary-limit methods using contraction and compactness. The uniqueness and attraction theorem in [12] uses a positive jump density bounded away from zero on compact subsets of $\mathbb{R}_+$. His arithmetic-jump example highlights the lattice phase. We construct an attracting periodic profile for both side distributions on the two-sided integer lattice. We then connect this profile to the slow community imbalance.

Sections 2–5 state and prove the results. Section 6 evaluates their

prediction steps. The supplement gives numerical certificates, statistical protocols, and provenance.

2 Model and main results

There are communities $C_0 = L_0 \cup R_0$ and $C_1 = L_1 \cup R_1$, each side of size m, and $q = 2m$. At each arrival choose an internal edge in C_s with probability $(1-\varepsilon)/2$, for each s, or direction L_0–R_1 or L_1–R_0 with probability $\varepsilon/2$ each. Internal edges are uniform on $K_{m,m}$. In cross direction d, a fixed nonnegative matrix $B^{(d)}$ has all row and column sums one; conditional edge probability is $B^{(d)}_{ij}/m$. Product sampling has $B = J/m$ and a matching is a permutation matrix. Matrices may depend deterministically on q. A randomly selected matrix is admissible if chosen independently before the arrivals, by conditioning on it.

Conditional on the fixed matrices, event types and candidate draws are independent across arrivals. Loads $A_v(n)$ are nonnegative integers, initially zero. Increment the endpoint of smaller load, breaking every tie to the left side. Let

$$M_s(n) = \sum_{v \in C_s} A_v(n), \quad \Delta_n = M_0(n) - M_1(n),$$
$$Z_c(n) = \sum_v (A_v(n) - c)_+.$$

For integer c, Z_c is also the number of overflows when each bucket stores its first c assigned items. Indeed the capped loads remain $\min(A_v, c)$ under the same candidate stream: below capacity both comparisons agree, and once both candidates are full the uncapped winner is irrelevant to the overflow count. The positive-part load functional therefore counts overflows.

For a pair of integer laws $z = (p_L, p_R)$ with finite first moments, write

$$x_k = p_L([k,\infty)), \quad y_k = p_R([k,\infty)),$$
$$a_k = x_k - x_{k+1}, \quad b_k = y_k - y_{k+1},$$

and $\mu(z) = (\mathbb{E}L + \mathbb{E}R)/2$. The metric d_1 is the sum of the two marginal Wasserstein distances, equivalently the sum of absolute tail differences. Integer shift S_j adds j to both laws.

There are three time scales. After k internal arrivals in one community, its mean clock is $u = k/q$. The common mean at global arrival n is $h_n = n/(2q)$. The slow diffusion clock is $t = n/q^2$. At critical fill $n_* = 2qc = 4mc$, these give $h_{n_*} = c$ and $n_*/q^2 = c/m \to \kappa$. An offset of $2hq$ arrivals changes the common mean by h while changing slow time by only $2h/q$. Fig. 2 shows the candidate structure and the two retained states.

For integer capacities c_q, set

$$n_* = 2q\, c_q,$$
$$n_q(h) = n_* + \lfloor 2q\, h \rfloor.$$

When $c_q/m \to \kappa > 0$, each bounded h window has nonnegative indices for large q. The imbalance path D_q linearly interpolates Δ_n/q at times n/q^2. The overflow curve F_q linearly interpolates $Z_{c_q}(k)/q$ at coordinates $h = (k - n_*)/(2q)$. Its distance from $Z_{c_q}(n_q(h))/q$ is at most $1/q$. The slow time associated with coordinate h is

$$\kappa_q = \frac{n_*}{q^2} = \frac{2c_q}{q}, \qquad h_q = \frac{n_q(h) - n_*}{2q},$$
$$\tau_q(h) = \frac{n_q(h)}{q^2} = \kappa_q + \frac{2h_q}{q}. \tag{2.1}$$

For bounded h, $h_q \to h$ and $\tau_q(h) \to \kappa$ uniformly. Thus the local overflow window collapses to one slow time, while h still resolves changes in fill level and lattice phase.

All path and curve Wasserstein distances below have order one. They use the uniform norm on the corresponding continuous-function space. Joint convergence uses the sum of the path and curve uniform norms. Thus, for a metric d, the transport distance is $\inf \mathbb{E}d(U,V)$ over couplings with the stated laws.

Theorem 2.1 (Coupling to product sampling)**.** Fix $T, \Lambda < \infty$ and assume $q \geq 4$, $\varepsilon \leq 1/4$ and $\varepsilon q \leq \Lambda$. For empty starts and any cross kernels specified above, the fixed-kernel and product processes admit a coupling such that, for $\mathcal{D}_n$ the sum of absolute differences of their four separately sorted side vectors,

$$\mathbb{E} \max_{n \leq Tq^2} \frac{\mathcal{D}_n}{q} = O_{T,\Lambda}(q^{-1/6} \log^2 q) \longrightarrow 0.$$

The same bound controls $|\Delta^F - \Delta^P|/q$ and $|Z^F_c - Z^P_c|/q$, simultaneously for all common capacities $c \geq 0$.

Theorem 2.2 (Internal periodic profile)**.** The mean-clock tail flow

$$\dot{x}_k = 2a_{k-1}y_{k-1}, \qquad \dot{y}_k = 2b_{k-1}x_k \tag{2.2}$$

has a unique finite-first-moment periodic family $\rho^u = (L_u, R_u)$ with mean u, satisfying

$$\rho^{u+1} = \mathsf{S}_1 \rho^u, \qquad \Phi_t \rho^u = \rho^{u+t}.$$

It has exponential moments, full support at every integer, and $d_1(\rho^u, \rho^v) = 2|u-v|$. Attraction is uniform over centered exponential-moment sublevels and all initial phases.

Define independent-draw comparisons and the overflow readout by

$$\begin{aligned} H(u,v) &= \mathbb{P}(L_u \leq R_v), \\ g(h,d) &= H(h + d/2, h - d/2) - H(h - d/2, h + d/2), \\ \overline{g}(d) &= \int_0^1 g(h,d)\, \mathrm{d}h, \\ \sigma(u) &= \tfrac{1}{2}\{\mathbb{E}[(L_u)_+] + \mathbb{E}[(R_u)_+]\}, \\ S_h(d) &= \sigma(h + d/2) + \sigma(h - d/2). \end{aligned} \tag{2.3, 2.4}$$

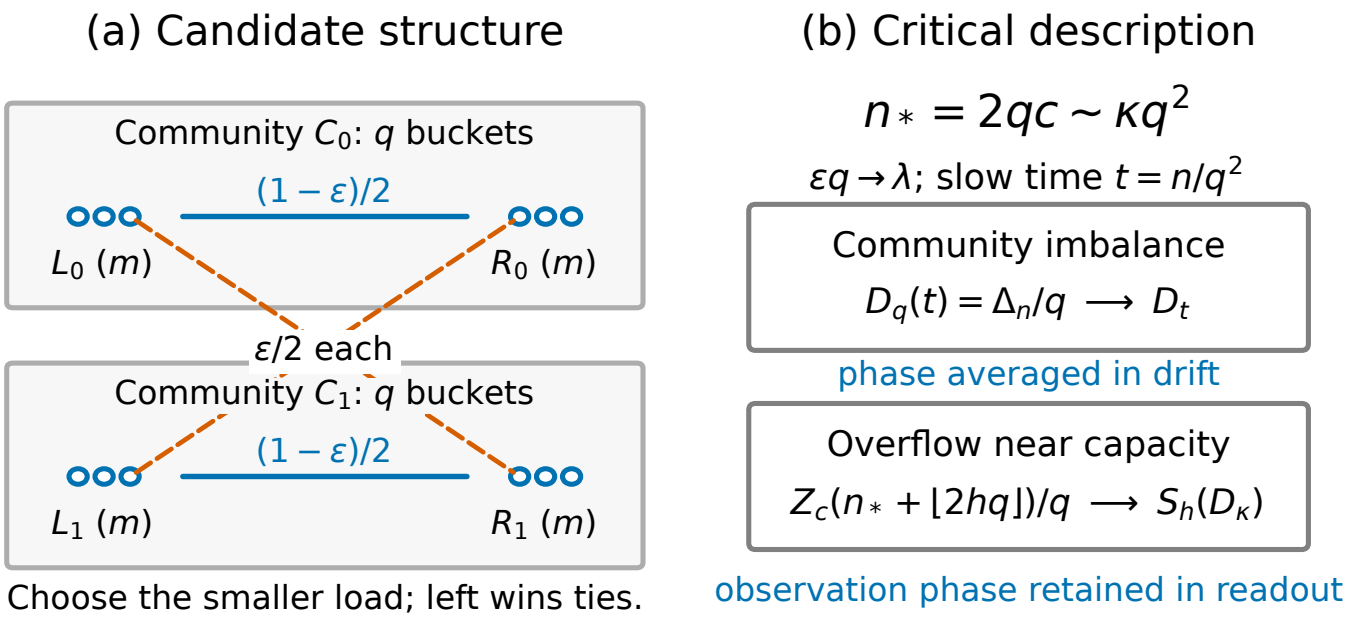


Fig. 2 The fixed allocation rule and its critical scales. Each of the four sides contains $m = q/2$ buckets. Internal directions have probability $(1-\varepsilon)/2$ each; cross directions have probability $\varepsilon/2$ each. The offered cross endpoints have uniform marginals under any fixed doubly stochastic kernel. The algorithm increments the smaller endpoint load and chooses the left endpoint at a tie. The limiting diffusion retains community imbalance; the overflow readout also retains the internal profile.

Theorem 2.3 (Joint critical limit)**.** Suppose $\varepsilon q \to \lambda \in [0,\infty)$, and let D solve

$$\mathrm{d}D_t = \mathrm{d}B_t + \lambda \overline{g}(D_t)\,\mathrm{d}t, \qquad D_0 = 0, \tag{2.5}$$

where B is standard Brownian motion.

Imbalance path. For every fixed T, $D_q \to D$ in first Wasserstein distance for the uniform path norm on $[0,T]$.

Critical overflow curve. Let the integer capacities satisfy $c_q/m \to \kappa \in (0,\infty)$. For each $H_0 < \infty$ and $T_* > \kappa$,

$$\left(D_q|_{[0,T_*]}, F_q|_{[-H_0,H_0]}\right) \longrightarrow \left(D|_{[0,T_*]}, (S_h(D_\kappa))_{|h|\le H_0}\right)$$

in first Wasserstein distance for the sum of the two uniform norms. The first coordinate retains the full diffusion path. The second component is the curve $h \mapsto S_h(D_\kappa)$, determined by the single terminal value D_κ at the critical slow time κ. Restriction gives joint convergence with any shorter path horizon.

The drift $\overline{g}$ is odd, smooth, strictly decreasing, globally 2-Lipschitz and negative on $(0,\infty)$. It satisfies $\overline{g}'(0) < 0$ and $\overline{g}(d) \to -\operatorname{sign}(d)$ at infinity. The function S_h is even and convex in d, 1-Lipschitz in d, and 2-Lipschitz in h, with $S_0(0) > 0$.

The drift averages the common phase along many complete cycles. The overflow readout retains the phase selected by h, including phase zero at the integer critical count. Section 5 proves the path limit by profile tracking and phase averaging. The curve estimate (5.12) then expresses F_q as a Lipschitz function of $D_q(\kappa)$ with vanishing expected uniform error. This gives their joint convergence.

Fig. 3 displays the profile and averaged response.

Corollary 2.4 (A leading internal overflow contribution)**.** Under Theorem 2.3, fix $\kappa > 0$ and $\lambda < \infty$. Let X_u have the equal mixture of the two laws in ρ^u, and set

$$r(u) = \mathbb{E}[(X_u)_+] - u_+ = \tfrac{1}{2}(\mathbb{E}|X_u| - |u|).$$

Then $r(u) > 0$ for every finite u, and

$$S_0(d) = \frac{|d|}{2} + r(d/2) + r(-d/2).$$

The mean omitted overflow satisfies

$$\lim_{q\to\infty} \mathbb{E}\left[\frac{Z_{c_q}(n_*) - |\Delta_{n_*}|/2}{q}\right] = \mathbb{E}\{r(D_\kappa/2) + r(-D_\kappa/2)\} > 0.$$

The positive limit may depend on κ and λ.

Proof. The mixture has mean u. Full support from Theorem 2.2 gives positive mass on both sides of zero. Hence $\mathbb{E}|X_u| > |\mathbb{E}X_u| = |u|$. Substitution into (2.4) gives the displayed decomposition.

At $n_* = 2qc_q$, the community excess totals are $\Delta_{n_*}/2$ and $-\Delta_{n_*}/2$. Their positive parts sum to $|\Delta_{n_*}|/2$. Also $F_q(0) = Z_{c_q}(n_*)/q$ and $D_q(\kappa_q) = \Delta_{n_*}/q$. Theorem 2.3 and the endpoint-time estimate (5.11) therefore give

$$(F_q(0), D_q(\kappa_q)) \longrightarrow (S_0(D_\kappa), D_\kappa)$$

in first Wasserstein distance.

Apply the Lipschitz function $(z,d) \mapsto z - |d|/2$ to this pair. Its expectation converges to $\mathbb{E}[S_0(D_\kappa) - |D_\kappa|/2]$. The displayed decomposition of S_0 identifies this expectation with the stated sum of r terms. Full support makes the integrand positive at every finite D_κ; its linear growth gives a finite expectation. □

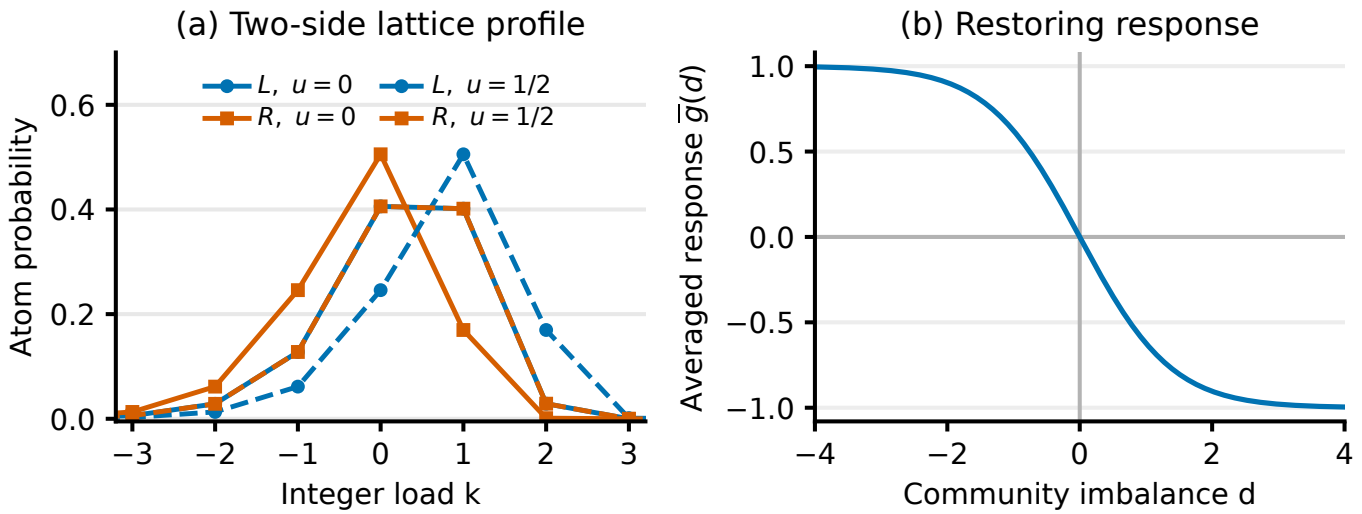


Fig. 3 Internal structure and restoring response, replotted from saved profile nodes and the response table. Left: left and right atom probabilities at mean phases $u = 0$ (solid) and $u = 1/2$ (dashed). The half-phase operation swaps the sides and shifts the new left law by one. Right: the phase-averaged response $\overline{g}(d)$; positive imbalance has negative response. Connecting segments join the saved nodes.

Corollary 2.5 (Capacity consequences)**.** Under Theorem 2.3, let $s_* = S_0(0) > 0$ and $C_{\kappa,\lambda} = \mathbb{E}S_0(D_\kappa)$. For each fixed $a \in (0, s_*)$,

$$\mathbb{P}\{Z_{c_q}(n_*) \geq aq\} \longrightarrow 1, \qquad \mathbb{P}\{Z_{c_q}(n_*) = 0\} \longrightarrow 0.$$

Moreover,

$$\mathbb{E}Z_{c_q}(n_*) = qC_{\kappa,\lambda} + o(q), \qquad \frac{\mathbb{E}Z_{c_q}(n_*)}{n_*} = \frac{C_{\kappa,\lambda}}{\kappa q} + o(q^{-1}).$$

Proof. Evenness and convexity give $S_0(d) \geq S_0(0) > a$ for every d. Weak convergence of $Z_{c_q}(n_*)/q$ and the open-set inequality therefore give the probability limits. For the mean, first Wasserstein convergence gives the bound

$$\left|\frac{\mathbb{E}Z_{c_q}(n_*)}{q} - C_{\kappa,\lambda}\right| \leq W_1\left(\mathcal{L}\left(\frac{Z_{c_q}(n_*)}{q}\right), \mathcal{L}(S_0(D_\kappa))\right) \longrightarrow 0.$$

Multiplication by q gives the leading mean asymptotic. Division by n_*, with $n_*/q^2 \to \kappa > 0$, gives the overflow fraction. □

Corollary 2.6 (Right-favored ties)**.** Theorems 2.1 and 2.3, and Corollaries 2.4 and 2.5, also hold when every tie favors the right endpoint. The limiting imbalance law and overflow readouts are unchanged. The internal periodic profile has its two side laws exchanged.

Proof. Exchange L_s and R_s in each community. A right-favored update becomes a left-favored update under this relabeling. Uniform internal sampling is preserved. The two cross matrices become $(B^{(1)})^\mathsf{T}$ and $(B^{(0)})^\mathsf{T}$, which remain fixed and doubly stochastic. The direction probabilities agree, so the relabeled process has the stated left-favored model law. Every community total and every overflow count is preserved pathwise. Apply the common critical law and exchange the internal side laws. □

■ 2.1 Capacity implications of aggregation and topology

Community totals can indicate balanced demand while individual buckets already overflow. Corollaries 2.4 and 2.5 quantify the positive order-q capacity cost hidden by this aggregate view at nominal fill. A capacity monitor can retain excess load and unused capacity together, or preserve the bucket-load histogram. The paired totals or histogram retain the within-community mismatch measured by equation (1.1).

Internal candidate access changes the size of this mismatch. In Section 6.5, cycles and disconnected matchings have mean omitted overflow 5.8 and 21.4 times the complete-bipartite mean at $q = 512$, $\kappa = \lambda = 1$, with left ties. For both graphs, the mean omitted overflow divided by q increases across the three tested sizes.

■ 3 From a fixed cross kernel to product sampling

We prove Theorem 2.1 by coupling the actual processes. Internal sampling first forgets a random label transposition under a dispersion bound. This controls the whole cross load-pair table. A sorted-rank coupling then uses greedy nonexpansion. Fix $\theta = 1/200$, $K_0 = 16/3$, and $I = (1 - \varepsilon)/2$.

Lemma 3.1 (Dispersion)**.** With $\xi_v = A_v - M_{s(v)}/q$ and $\Gamma = \sum_v \cosh(\theta\xi_v)$, the centered potential satisfies

$$\sup_{n\geq 0} \mathbb{E}\Gamma_n \leq K_0 q. \tag{3.1}$$

Proof. The rank-based potential argument is given in Supplement section SM2.1. It uses only the cross marginals. □

Lemma 3.2 (Random transposition)**.** Let μ_n be the law of the actual fixed-kernel configuration. Independently choose two uniform labels i, j in one side, and let τ swap them. For $\ell \geq 4$, integer $L = q\ell$, and

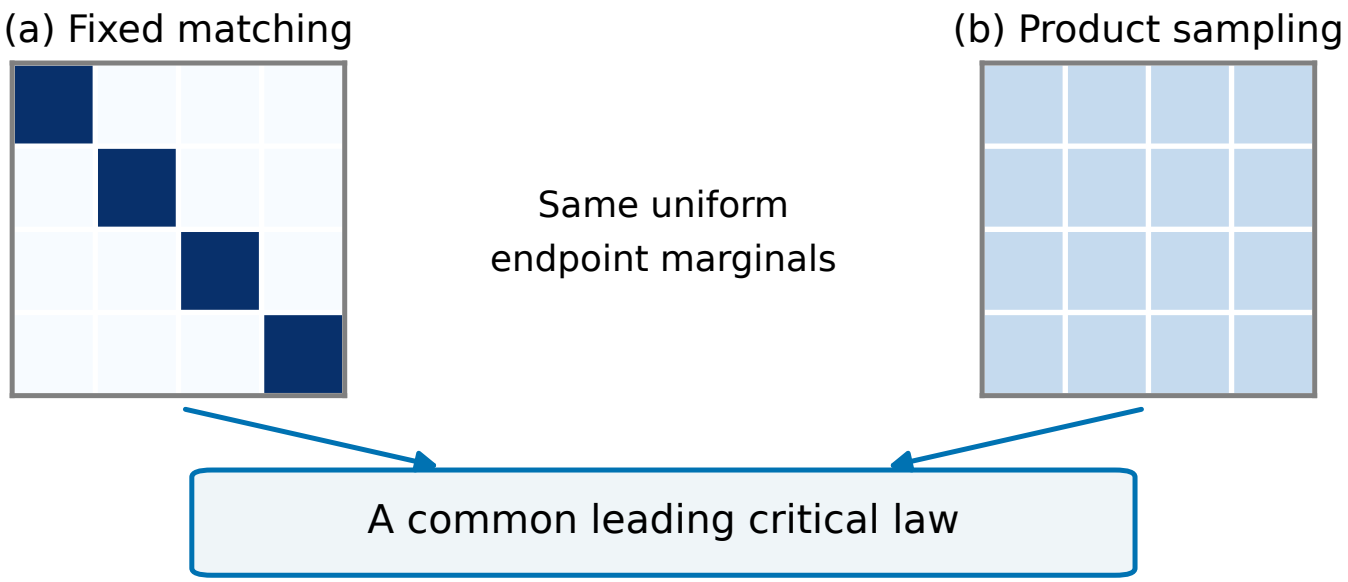


Fig. 4 Cross-candidate distributions for matching and product sampling. The grids show conditional candidate-pair probabilities in one cross direction, with four illustrative labels on each side. Matching probabilities are $1/4$ on the diagonal and zero elsewhere; product probabilities are all $1/16$. Both have uniform marginals. Internal sampling and rare cross arrivals support the process comparison.

$n \geq L$,

$$\mathbb{E}_{i,j}\,\mathrm{TV}(\mu_n, \tau\mu_n) \leq \eta, \tag{3.2}$$

$$\eta = \min\left\{1, \frac{32K_0}{\theta\sqrt{\ell}} + 16K_0 e^{-\theta\ell/16} + e^{-L/48} + 4\varepsilon\ell\right\}.$$

Proof. Compare the original and transposed configurations through the conjugate kernel. The two discrepancies behave as tags until they meet. Suppress cross events touching a tag; internal updates then make the tag gap no harder to absorb than a fair walk. Dispersion controls the initial gap and deviations at the window endpoints. On the remaining event, nonmeeting requires at least $\lceil \ell/4 \rceil$ embedded jumps. A fixed-jump reflection bound gives the first term of η. The conjugate-kernel comparison adds a second cross-touch allowance. Supplement section SM2.4 gives the full coupling and constants. □

For one cross direction write Q_n for its actual conditional joint load-pair law and p_n, r_n for the empirical side marginals. If $0 \leq f \leq 1$, define

$$C_B(f) = \frac{1}{m}\sum_{ij} B_{ij} f(U_i, V_j) - \frac{1}{m^2}\sum_{ij} f(U_i, V_j).$$

The test may use parameters unchanged by the side swap, including community means. Let I_0 be uniform, draw $J_0 \mid I_0$ from the corresponding row of B, and draw J_0' independently and uniformly. We claim

$$\mathbb{E}C_B(f)^2 \leq 2\eta + 2/m, \qquad \mathbb{E}|C_B(f)| \leq \sqrt{2\eta + 2/m}. \tag{3.3}$$

Write $C = C_B(f)$ and take these auxiliary indices independently of the configuration. Pointwise,

$$C^2 = \mathbb{E}_{\mathrm{aux}}\left[C\{f(U_{I_0}, V_{J_0}) - f(U_{I_0}, V_{J_0'})\}\right].$$

For fixed indices let τ exchange J_0, J_0'. The product-marginal term of C is unchanged, and $|C(\tau A) - C(A)| \leq 2/m$. Apply total variation to $G(A) = C(A) f(U_{I_0}, V_{J_0})$, for which $|G| \leq 1$. Thus the expected difference is at most $2\,\mathrm{TV}(\mu_n, \tau\mu_n) + 2/m$. After averaging, (J_0, J_0') are independent uniform side labels: column sums make J_0 uniform, even though it depends on I_0. Lemma 3.2 gives (3.3). The bounded-test TV cost is 2η.

Center each load coordinate at its community's integer mean part. For each fixed offset cell in $[-R, R]^2$, its indicator satisfies (3.3). Summing over this deterministic square and using (3.1) outside it yields

$$\mathbb{E}\,\|Q_n - p_n \otimes r_n\|_1 \leq \rho,$$
$$\rho = \min\{2, (2R+1)^2\sqrt{2\eta + 2/m} + 16K_0 e^{-\theta(R-1)}\}. \tag{3.4}$$

Indeed Q and the product have the same marginals; twice their marginal tail masses bounds the mass outside the square. Replacing the mean by its integer part costs at most one in the exponential bound.

It remains to compare two actual process laws. Sort each side separately. Integer greedy updates using the same offered ranks are nonexpansive in sorted ℓ_1 distance: if different winners are chosen, the two relevant coordinate differences have the ordering needed for their $+1, -1$ changes not to increase the absolute sum; sorting again minimizes the matching cost. Internally use common independent uniform ranks. At a cross, sample the actual fixed edge in the fixed process, then maximally couple its load pair from Q_F to a virtual pair from $p_F \otimes r_F$. Given the virtual loads, choose ranks independently and uniformly inside the equal-load blocks. Conditional on the entire joint past before drawing the current edge, these ranks are independent uniforms, since a block of size k has mass k/m followed by probability $1/k$ per rank. Use the same ranks in the product process, which therefore has the required product transition law.

On successful load-pair coupling the actual and virtual sorted fixed-process updates agree, including ties. On failure their distance is at most two. Hence

$$\mathcal{D}_{n+1} \le \mathcal{D}_n + 2\mathbf{1}_{\{\text{cross and load-pair coupling fails}\}}.$$

Before L use failure probability one; afterwards its expected conditional value is at most $\rho/2$. Summing the nonnegative counts gives the explicit bound

$$\mathbb{E}\max_{n\le N}\mathcal{D}_n \le 2\varepsilon\min(N,L) + \varepsilon\rho(N-L)_+. \tag{3.5}$$

Take $\ell = \max(4, \lceil q^{2/3}\rceil)$ and $R = \lceil (2/\theta)\log q\rceil + 1$. Then (3.5) divided by q, for $N \le Tq^2$, is $O(q^{-1/6}\log^2 q)$. Both the community difference and every positive-part sum are 1-Lipschitz in sorted load distance. This proves Theorem 2.1.

■ 4 The mean-locked internal orbit

To prove Theorem 2.2, we first write the difference of two tail flows as a positive, mass-preserving midpoint evolution. Its ℓ_1 norm cannot increase (4.2). At equal means, positive and negative differences reach a common state, giving the strict inequality (4.3). On each mean slice, moment compactness gives a unique fixed point of the one-period map. A finite-cover argument upgrades pointwise attraction to the uniform estimate (4.4).

In this section all times use the mean clock. Signed lattice coordinates express translated laws; actual allocations remain nonnegative. Equation (2.2) has a unique probability flow on finite-first-moment laws. It can be constructed by two coupled bounded-rate birth laws with left rate $2y_k$ and right rate $2x_{k+1}$. Picard iteration, or finite truncation followed by tightness, gives existence and uniqueness. The tail field is globally Lipschitz, probability constraints are invariant, and rates at most two prevent explosion. The sum of the mean derivatives is two, so $\mu(\Phi_t z) = \mu(z) + t$.

■ 4.1 Moment compactness and exact nonexpansion

Let

$$V_\theta(z) = \mathbb{E}\cosh(\theta(L-\mu)) + \mathbb{E}\cosh(\theta(R-\mu)).$$

The selected-load distribution is $w_k = a_k y_k + b_k x_{k+1}$, of total mass one, bounded by twice the pooled law $\nu = (p_L + p_R)/2$. Its suffix is $x_k y_k \le [(x_k + y_k)/2]^2$. The same rank/Gini argument as in Lemma 3.1, now with pure-internal rank deficit one, and differentiation of the moving mean give

$$\frac{\mathrm{d}}{\mathrm{d}t}V_\theta(\Phi_t z) \le -\{\theta/2 - 2(e^\theta - 1 - \theta)\}V_\theta + \theta \le -\theta V_\theta/4 + \theta. \tag{4.1}$$

Truncation justifies this calculation for finite exponential moments. Thus every centered sublevel $V_\theta \le M$, $M \ge 4$, is invariant. The moment bound supplies the invariant sublevels used in the attraction argument.

For two solutions put $d_k = x_k - \widetilde{x}_k$, $e_k = y_k - \widetilde{y}_k$, and let bars denote midpoint tails or atoms. Their difference satisfies exactly the forward equation on states X_k, Y_k with transitions

$$X_k \longrightarrow X_{k+1} \quad \text{at rate } 2\bar{y}_k, \qquad X_k \longrightarrow Y_k \quad \text{at rate } 2\bar{b}_{k-1},$$
$$Y_k \longrightarrow Y_{k+1} \quad \text{at rate } 2\bar{x}_{k+1}, \qquad Y_k \longrightarrow X_{k+1} \quad \text{at rate } 2\bar{a}_k.$$

Exit rates are $2\bar{y}_{k-1}$ and $2\bar{x}_k$, at most two. The midpoint identity is exact for the bilinear field. Its evolution is positive and mass preserving, so

$$d_1(\Phi_t z, \Phi_t \widetilde{z}) \le d_1(z, \widetilde{z}). \tag{4.2}$$

It also shows that the field is 4-Lipschitz in tail ℓ_1.

■ 4.2 Strictness on an equal-mean slice

Supplement section SM8.2 derives explicit lower bounds $Ct^p e^{-\gamma t}$ for all sufficiently high atoms, starting from one positive atom on each side. Consequently all finite tails are positive for $t > 0$. On a compact subinterval of $(0, t)$, any two finite midpoint states can follow finite paths to a common high state X_K. The ordered-jump bound in Supplement section SM8.3 makes both transition probabilities $P_{0,t}(i, X_K), P_{0,t}(j, X_K)$ positive. These bounds depend on the flows and chosen states.

For equal means the initial signed tail difference $v = (d, e)$ has total mass zero. If $v \ne 0$, choose $v_i > 0$ and $v_j < 0$. Positivity and mass preservation of the transition kernel give the explicit cancellation bound

$$\begin{aligned}\|v(t)\|_1 &\le \|v(0)\|_1 \\ &\quad - 2\min\{v_i P_{0,t}(i, X_K), (-v_j)P_{0,t}(j, X_K)\} \\ &< \|v(0)\|_1.\end{aligned}$$

Thus for every $t > 0$,

$$d_1(\Phi_t z, \Phi_t \widetilde{z}) < d_1(z, \widetilde{z}) \quad \text{if } \mu(z) = \mu(\widetilde{z}),\ z \ne \widetilde{z}. \tag{4.3}$$

■ 4.3 Existence, uniqueness and uniform attraction

Define $T = \mathsf{S}_{-1}\Phi_1$ and $C_{\alpha,M} = \{\mu = \alpha, V_\theta \le M\}$, $M \ge 4$. Exponential tails and lower semicontinuity make this set compact in d_1. Identical mixtures on $\{0, 1\}$ show nonemptiness for $\alpha \in [0, 1]$. The moment estimate shows that T preserves each slice. Supplement sections SM8.1 and SM8.4 give the full details of these steps and the following fixed-point argument.

Along an orbit, $d_1(T^n z, T^{n+1} z) \downarrow r$. A convergent subsequence with limit w gives $d_1(w, Tw) = d_1(Tw, T^2 w) = r$. Strictness forces $Tw = w$ and $r = 0$. Two fixed points contradict strictness. For any orbit, a subsequential limit of its decreasing distance to w would give the same equality unless that distance tends to zero. Thus every orbit converges to the unique fixed point on its slice.

Let z_* be the fixed point in $C_{0,4}$ and define

$$\rho^u = \mathsf{S}_{\lfloor u\rfloor}\Phi_{\{u\}} z_*.$$

The fixed-point identity gives the flow and period identities, continuity across integer phases, and $V_\theta(\rho^u) \le 4$. Strict contraction gives uniqueness among periodic families with finite first moments.

For fixed $M \ge 4$, the set $K_M = \{z : \mu(z) \in [0, 1], V_\theta(z) \le M\}$ is compact. The functions $f_n(z) = d_1(T^n z, \rho^{\mu(z)})$ are continuous, decrease pointwise to zero, and satisfy $f_{n+1} \le f_n$ by nonexpansion. For $\delta > 0$, the increasing open sets $\{f_n < \delta\}$ cover K_M. A finite subcover

has a largest index N; hence $\sup_{K_M} f_n \le \delta$ for $n \ge N$. Nonexpansion for the fractional part of time then gives

$$\omega_M(t) := \sup_{\substack{\mu(z)\in[0,1]\\ V_\theta(z)\le M}} d_1(\Phi_t z, \rho^{\mu(z)+t}) \longrightarrow 0. \tag{4.4}$$

Integer translation covers arbitrary means. Along the birth flow each side increases stochastically. Summing the two mean increments therefore gives

$$d_1(\rho^u, \rho^v) = 2|u - v|. \tag{4.5}$$

For any integer k, positivity at $k + n$ after a sufficiently large integer time n transfers to the atom at k by the period relation. This gives full support and completes Theorem 2.2.

4.4 Interlacing, a single front, and the response

The empty-start flow has invariant inequalities $x_k \ge y_k \ge x_{k+1}$. At a boundary equality its gap derivative is nonnegative, directly from (2.2); truncation extends the check to the full lattice. Attraction of centered integer-time empty profiles transfers the inequalities to the orbit. The map $J(p_L, p_R) = (\mathsf{S}_1 p_R, p_L)$ commutes with the flow, has $J^2 = \mathsf{S}_1$, and raises the mean by $1/2$. Uniqueness implies

$$\rho_L^{u+1/2} = \mathsf{S}_1 \rho_R^u, \qquad \rho_R^{u+1/2} = \rho_L^u.$$

Set $z_{2k} = x_k$, $z_{2k+1} = y_k$. The equations and half-phase identity yield a single normalized front F:

$$\begin{aligned} x_k(u) = F(2k - 2u), \quad y_k(u) = F(2k + 1 - 2u),\\ -F'(r) = F(r - 1)[F(r - 2) - F(r)]. \end{aligned} \tag{4.6}$$

It decreases strictly from one to zero, is smooth by repeated differentiation, and its translation is fixed by the mean-zero normalization.

The function $w(r) = [F(r) - F(r + 2)]/2$ is a positive density of mass one. Expanding H as its atom–tail sum and substituting $r = 2k - 2h - d$ in each term gives

$$\overline{g}(d) = \int_{\mathbb{R}} w(r)[F(r + 1 + 2d) - F(r + 1 - 2d)]\,\mathrm{d}r. \tag{4.7}$$

The phase intervals tile the real line; Tonelli applies to the original nonnegative sums. Equation (4.7) proves oddness, strict decrease and the far-field limit by dominated convergence. Bounded derivatives from (4.6) give smoothness. Marginal optimal couplings and (4.5) give the useful global estimate

$$|g(h, d) - g(h', d')| \le 4|h - h'| + 2|d - d'|. \tag{4.8}$$

An integer comparison changes by at most the integer displacement.

Differentiating the absolutely convergent atom sums gives

$$\partial_1 H(u, u) = -2 \sum_k a_k y_k b_k, \qquad \partial_2 H(u, u) = 2 \sum_k a_k x_k b_{k-1}.$$

Integration of the derivative of periodic $H(u, u)$ equates the two sums after phase integration. Thus the restoring coefficient is

$$\alpha = -\overline{g}'(0) = 4 \int_0^1 \sum_k a_k(u) y_k(u) b_k(u)\,\mathrm{d}u > 0. \tag{4.9}$$

The front form is $\alpha = -4 \int w(r) F'(r+1)\,\mathrm{d}r$. Finally the orbit moment bound implies, by a union bound on centered deviations,

$$|\overline{g}(d) + \mathrm{sign}(d)| \le \min(2, 32e^{-\theta|d|/2}), \qquad d \ne 0.$$

5 Tracking, deterministic averaging and critical overflow

The proof of Theorem 2.3 has three steps. Deterministic restarts combine fixed-window fluid approximation with profile attraction, giving the tracking estimate (5.5). The imbalance martingale supplies Brownian noise; blocks of $2q$ arrivals average its drift, giving the remainder estimate (5.8). Finally, positive-part Lipschitz continuity and a finite mesh give the curve estimate (5.12). The curve estimate identifies the limit through the same terminal imbalance, giving joint convergence.

We work with product cross sampling until the final coupling transfer. Discarding finitely many q, choose $\Lambda < \infty$ such that $\varepsilon q \le \Lambda$ and $\varepsilon \le 1/4$.

5.1 A uniform estimate on a fixed fluid window

Consider one pure community, with $q = 2m$ buckets. Let X_k^q be its pair of empirical side laws after k internal arrivals. The mean clock is $u = k/q$. Interpolate linearly between these times. Start the fluid $X(u) = \Phi_u X_0^q$ from the exact empirical laws. Assume

$$V_\theta(X_0^q) \le R, \qquad \theta = 1/200.$$

Translate all loads by $-\lfloor \mu(X_0^q) \rfloor$. This leaves the metric and dynamics unchanged. The translated initial mean belongs to $[0, 1)$. Fix an integer $B \ge 1$ and a cutoff $K_1 \ge 1$. Then

$$\mathbb{E} \sup_{0\le u\le B} d_1(X^q(u), X(u)) \le \eta_q(B, R, K_1). \tag{5.1}$$

The bound is uniform over all empirical initial states satisfying the displayed moment condition.

One explicit choice is

$$\begin{aligned} \eta_q &= e^{4B}\big[8(2K_1 + 1)\sqrt{B/q}\\ &\qquad + 4BC_{R,B}e^{-\theta(K_1-1)} + 8B/q\big],\\ C_{R,B} &= 2e^\theta R \exp\{2B(e^\theta - 1)\}. \end{aligned}$$

Subtracting the conditional tail increments produces a martingale. Doob's inequality bounds each retained coordinate by $4\sqrt{B/q}$ in expected maximum; the exponential moment bounds the omitted tails. The field is 4-Lipschitz, so Gronwall gives the displayed bound, including the interpolation error $8B/q$. Supplement section SM2.2 details these estimates. With

$$K_1(q) = \lceil 2\log(q)/\theta \rceil,$$

the bound tends to zero for each fixed B, R.

5.2 Moments, cross suppression, and deterministic restarts

For the real product process, let W_n be the sum of four side distances to orbit laws at the actual community means. Let Ψ_n sum the four empirical cosh moments centered at these means. The required bounds are

$$\sup_n \mathbb{E}\Psi_n \le C_E = 32/3, \qquad \sup_n \mathbb{E}W_n^2 \le C_*^2 = (32C_E + 128)/\theta^2. \tag{5.2}$$

For the first bound, $\Psi_n = 2\Gamma_n/q$, so apply Lemma 3.1. For the second, let J_n sum the four empirical mean absolute deviations from their community means. Jensen's inequality gives

$$e^{\theta J_n/4} \le \Psi_n/2, \qquad \mathbb{E}J_n^2 \le 16C_E/\theta^2.$$

The four orbit deviations sum to at most $8/\theta$ by the profile moment bound. Hence $W_n \le J_n + 8/\theta$; squaring proves (5.2).

Start an auxiliary chain from the real state at a deterministic time. Use the same internal candidates, but suppress every cross update. Greedy integer updates with the same candidates are nonexpansive in labelled ℓ_1. If $N_\times$ crosses occur during a window, the total labelled discrepancy is at most $N_\times$. Normalizing each side by $m = q/2$ gives profile distance at most $2N_\times/q$. Each suppressed cross removes one unit of community mass. The sum of the two mean discrepancies is therefore $N_\times/q$. Equation (4.5) adds at most another $2N_\times/q$ for the two orbit targets. Consequently

$$W_n \le W_n^{\mathrm{aux}} + 4N_\times/q. \tag{5.3}$$

Fix $\delta > 0$ and $R \ge 4$. Choose a fixed integer $A \ge 1$ such that $\omega_R(u) \le \delta/2$ for all $u \ge A$. Put

$$L = 2q(A+1), \qquad B = 2(A+1)+1.$$

For any deterministic endpoint $n \ge L$, restart at $n - L$. A start is good if $\Psi_{n-L} \le R$. Its complement has probability at most C_E/R.

Let N_s be the internal event count for community s in the window. Its marginal law is $\mathrm{Bin}(L, (1-\varepsilon)/2)$. For sufficiently large q, its mean exceeds $Aq + q/2$, while its variance is at most $L/4$. Chebyshev and a union bound give

$$\mathbb{P}\{\min(N_0, N_1) < Aq\} \le C_A/q, \qquad C_A = 4(A+1).$$

Also $N_s \le L < Bq$. The union bound controls the two clocks jointly. The internal candidate streams can be chosen independently of all event-type clocks and the restart state. On good starts, the supremum in (5.1) therefore covers every terminal clock.

On good starts and clocks, the auxiliary fluid targets are within total distance δ of the orbit. The two empirical approximation errors contribute at most $2\eta_q(B, R, K_1)$. On bad starts or clocks, use Cauchy–Schwarz for the real W_n, with (5.2). Taking expectations in (5.3) yields, uniformly over deterministic $n \ge L$,

$$\mathbb{E}W_n \le \delta + 2\eta_q(B, R, K_1) + 4\varepsilon L/q + C_*\sqrt{C_E/R} + C_*\sqrt{C_A/q}. \tag{5.4}$$

First fix δ, R, then choose A, B. Next let $q \to \infty$. Finally let $R \to \infty$ and $\delta \to 0$. This proves

$$\lim_{H\to\infty} \limsup_{q\to\infty} \sup_{n \ge Hq} \mathbb{E}W_n = 0, \qquad q^{-2} \sum_{n < Tq^2} \mathbb{E}W_n \longrightarrow 0. \tag{5.5}$$

For the second conclusion, the first Hq terms contribute at most C_*H/q. The remaining terms are bounded by $T \sup_{n \ge Hq} \mathbb{E}W_n$. In particular, $\mathbb{E}W_{n_q} \to 0$ whenever the deterministic times satisfy $n_q/q^2 \to t > 0$.

■ 5.3 Exact imbalance decomposition and Brownian noise

Let $\mathcal{F}_n$ contain the history before arrival $n+1$. Write $D_n^q = \Delta_n/q$, $h_n = n/(2q)$, and $\lambda_q = \varepsilon q$. The exact community means are

$$M_0(n)/q = h_n + D_n^q/2, \qquad M_1(n)/q = h_n - D_n^q/2.$$

For independent product-cross candidates, define

$$g_n^{\mathrm{act}} = \mathbb{P}(L_0 \le R_1 \mid \mathcal{F}_n) - \mathbb{P}(L_1 \le R_0 \mid \mathcal{F}_n).$$

Every increment $\xi_{n+1} = \Delta_{n+1} - \Delta_n$ is ± 1. The internal contributions to its mean cancel. The two cross directions give

$$b_n = \mathbb{E}(\xi_{n+1} \mid \mathcal{F}_n) = \varepsilon g_n^{\mathrm{act}}, \qquad |b_n| \le \varepsilon.$$

Define the discrete martingale $\mathcal{M}_k = q^{-1}\sum_{n<k}(\xi_{n+1} - b_n)$. Apply the martingale functional central limit theorem (FCLT) to the step process $\mathcal{M}_{\lfloor tq^2 \rfloor}$. Its maximum jump is at most $2/q$, and

$$\langle \mathcal{M} \rangle_q(t) = q^{-2} \sum_{n < \lfloor tq^2 \rfloor} (1 - b_n^2), \quad \sup_{t \le T} |\langle \mathcal{M} \rangle_q(t) - t| \le q^{-2} + T\varepsilon^2.$$

Bracket jumps are at most q^{-2}. These deterministic bounds verify the three conditions of [14, Theorem 2.1(ii)], including the expected squared-jump condition. Thus the step martingale converges to standard Brownian motion. Let M_q denote its linear interpolation. The interpolation changes it by at most $2/q$. Since the limit is continuous, M_q converges weakly in $C([0,T])$ with its uniform norm. Discrete Doob also gives

$$\mathbb{E}\|M_q\|_{\infty,[0,T]}^2 \le 4(T + q^{-2}). \tag{5.6}$$

■ 5.4 Deterministic averaging of the common phase

Integer optimal couplings give

$$|g_n^{\mathrm{act}} - g(h_n, D_n^q)| \le W_n.$$

Indeed, for integer arguments an indicator $\mathbf{1}_{\{l \le r\}}$ can change by at most the sum of the integer displacements. Couple each of the four side marginals independently with its orbit target. Equations (5.5) and $\lambda_q \le \Lambda$ imply

$$\mathbb{E} \sup_{t \le T} \left| \frac{\varepsilon}{q} \sum_{n < \lfloor tq^2 \rfloor} \{ g_n^{\mathrm{act}} - g(h_n, D_n^q) \} \right| \le \Lambda q^{-2} \sum_{n < Tq^2} \mathbb{E}W_n \longrightarrow 0.$$

The function $g(h, d)$ is one-periodic in h, by integer translation of both orbit laws. Partition arrival indices into blocks of length exactly $2q$. For a block starting at j, discrete Doob and $|b_n| \le \varepsilon$ give

$$\mathbb{E} \max_{0 \le r \le 2q} |D_{j+r}^q - D_j^q| \le 2\sqrt{2/q} + 2\varepsilon.$$

Freeze D_n^q at the block start. The phases $h_n \bmod 1$ cover the $2q$ equally spaced points of one period. Equation (4.8) bounds their Riemann-sum error by $2/q$, uniformly in the frozen imbalance. The Lipschitz constants of g and $\overline{g}$ in d are both at most two. Freezing and unfreezing

therefore cost at most four times the displayed block variation. A final incomplete block contributes at most $4/q$ after normalization. Hence

$$\mathbb{E}\sup_{t\le T}\left|q^{-2}\sum_{n<\lfloor tq^2\rfloor}[g(h_n,D_n^q)-\overline{g}(D_n^q)]\right| \le C(T+1)(q^{-1/2}+\varepsilon+q^{-1})\longrightarrow 0. \tag{5.7}$$

The common phase traverses one period every $2q$ arrivals.

Let D_q be the linear interpolation of D_n^q at times n/q^2. Combining the last two estimates, the exact martingale decomposition, and a Riemann sum gives

$$D_q(t)=M_q(t)+\lambda\int_0^t\overline{g}(D_q(s))\,\mathrm{d}s+R_q(t),\qquad \mathbb{E}\|R_q\|_{\infty,[0,T]}\longrightarrow 0. \tag{5.8}$$

The remaining errors are bounded by $T|\lambda_q-\lambda|$, $O_T(\Lambda/q)$ for interpolation inside the drift integral, and $O(1/q)$ for fractional final steps. These bounds use $|\overline{g}|\le 1$ and its global Lipschitz constant two.

For a continuous driver m, let $\Gamma_\lambda(m)$ solve

$$x(t)=m(t)+\lambda\int_0^t\overline{g}(x(s))\,\mathrm{d}s.$$

Picard iteration gives existence and uniqueness. Gronwall gives

$$\|\Gamma_\lambda(m)-\Gamma_\lambda(\widetilde{m})\|_\infty\le e^{2\lambda T}\|m-\widetilde{m}\|_\infty,$$
$$\|D_q-\Gamma_\lambda(M_q)\|_\infty\le e^{2\lambda T}\|R_q\|_\infty.$$

The weak convergence of M_q, together with (5.6), implies first Wasserstein convergence. For example, use a Skorokhod representation in the Polish space $C([0,T])$. Uniform second moments make the represented uniform distances uniformly integrable. Their almost sure convergence to zero then gives convergence in mean. Applying the Lipschitz map Γ_λ and (5.8) proves first Wasserstein convergence of D_q to the solution of (2.5). This also identifies Brownian variance one.

The other drift claims in Theorem 2.3 follow from (4.7). The density w has mass one and is strictly positive. The front F is smooth and strictly decreasing, with bounded derivatives of every fixed order. Differentiation under the integral is therefore valid. In particular,

$$\overline{g}'(d)=2\int_{\mathbb{R}}w(r)\{F'(r+1+2d)+F'(r+1-2d)\}\,\mathrm{d}r<0.$$

Oddness, negativity for $d>0$, and the far-field sign limits follow directly. Equation (4.9) gives $-\overline{g}'(0)>0$. The marginal coupling estimate gives the global Lipschitz bound.

■ 5.5 The endpoint overflow curve

The positive-part functional along the orbit satisfies

$$\sigma'(u)=\sum_{k\ge 0}\{a_ky_k+b_kx_{k+1}\}=x_0(u)y_0(u). \tag{5.9}$$

The sum telescopes. Equivalently, a positive-part increment occurs when both offered loads are nonnegative. The tails increase with u. Thus σ is convex, nondecreasing, and 1-Lipschitz. Equation (2.4) gives evenness and convexity in d, the stated 1-Lipschitz bound in d, and the 2-Lipschitz bound in h. Full support gives $S_0(0)>0$.

The numerical error budget uses a sharper bound in d. Since $0\le\sigma'\le 1$, differentiation gives

$$\partial_dS_h(d)=\tfrac{1}{2}\{\sigma'(h+d/2)-\sigma'(h-d/2)\},\qquad |\partial_dS_h(d)|\le\tfrac{1}{2}.$$

Thus S_h is $1/2$-Lipschitz in d.

Fix $H_0<\infty$ and write $\mathcal{S}(d)=(S_h(d))_{|h|\le H_0}$ for the overflow curve. Write $\kappa_q=n_*/q^2=2c_q/q\to\kappa>0$ and choose a path horizon $T_*>\kappa$. Increase T_* if a longer joint path is requested. All window times belong to $[0,T_*]$ for large q.

Define the continuous curve F_q by linear interpolation of

$$F_q\left(\frac{k-n_*}{2q}\right)=\frac{Z_{c_q}(k)}{q}.$$

It is deterministically 2-Lipschitz on $[-H_0,H_0]$. Its distance from the original floor-indexed curve is at most $1/q$. This specifies the interpolation in Theorem 2.3.

At a fixed h, put $n=n_q(h)$, $h_q=(n-n_*)/(2q)$, and $\tau_q(h)=n/q^2$. Then $|h_q-h|\le 1/(2q)$. Integer translation by c_q and positive-part Lipschitz continuity give the comparison

$$\left|\frac{Z_{c_q}(n)}{q}-S_{h_q}(D_q(\tau_q(h)))\right|\le W_n/2. \tag{5.10}$$

The factor $1/2$ comes from $m/q=1/2$ for each side. Since $\tau_q(h)\to\kappa>0$, the expectation of W_n tends to zero by (5.5).

We also need to move the slow endpoint from $\tau_q(h)$ to κ. Put

$$a_q=|\kappa_q-\kappa|+2H_0/q+2/q^2\longrightarrow 0.$$

All these times lie within a_q of κ. A martingale oscillation bound on that deterministic interval, followed by the bounded drift estimate, gives

$$\mathbb{E}\sup_{|h|\le H_0}|D_q(\tau_q(h))-D_q(\kappa)|\le C\sqrt{a_q+q^{-2}}+C\Lambda a_q+C/q\longrightarrow 0. \tag{5.11}$$

For this bound, start the martingale at the left endpoint of the interval. Doob controls its maximum oscillation by the square root of the interval length. Taking differences through that endpoint controls every requested pair of times.

Let $G_q(h)=S_h(D_q(\kappa))$. Combining (5.10), (5.11), and the interpolation bounds gives

$$\mathbb{E}|F_q(h)-G_q(h)|\longrightarrow 0\quad\text{for every fixed } h.$$

Both F_q and G_q are 2-Lipschitz. For any fixed mesh with maximum gap at most η,

$$\|F_q-G_q\|_\infty\le\max_{h\text{ in the mesh}}|F_q(h)-G_q(h)|+4\eta.$$

Take expectations, then $q\to\infty$, then $\eta\downarrow 0$. The finite maximum is bounded by the sum of its errors. We obtain the same-process estimate

$$\mathbb{E}\|F_q-\mathcal{S}(D_q(\kappa))\|_{\infty,[-H_0,H_0]}\longrightarrow 0. \tag{5.12}$$

The finite mesh upgrades the deterministic-time estimates to uniform control of the overflow curve.

The map $f \mapsto (f, \mathcal{S}(f(\kappa)))$ is Lipschitz from the uniform path norm to the sum of the two uniform norms. First Wasserstein convergence of D_q and (5.12) therefore prove the claimed joint path-and-curve convergence. Restriction gives the result on any shorter requested path horizon. Finally, apply Theorem 2.1 to couple the fixed-kernel and product processes on $[0, T_* q^2]$. The joint path-and-curve transport cost is at most twice $\max_n \mathcal{D}_n/q$, up to vanishing interpolation errors. Its expectation tends to zero. This transfers the first Wasserstein conclusions and finishes Theorem 2.3.

6 Evaluation of the critical law

We evaluate the internal profile at observed imbalance, then compare the terminal diffusion and unconditional overflow mean with microscopic simulations. Section 6.5 tests how ties and internal candidate graphs affect the omitted overflow.

6.1 Numerical evaluation

We compute the profile readout using an a posteriori certificate based on a saved probability trajectory. Binary64 inputs are treated as exact rationals. The certificate combines a full polynomial residual, analytic infinite-tail bounds, and lower transition kernels for the midpoint difference equation. The uniform profile error is $D_* < 9.038\,10^{-6}$. The numerical bounds are

$$S_0(0) \in [0.63127654,\ 0.63129462]$$

and $-\overline{g}'(0) \in [0.71997987,\ 0.72012468]$. Supplement sections SM3.1 and SM4 give the construction and all reported readout bounds.

Fig. 3 shows the two-side profile and averaged response from the saved numerical nodes. The different side laws record the tie rule. The half-phase shift agrees with the identity proved in Section 4. The response points toward zero imbalance and saturates at large imbalance.

6.2 The internal contribution at observed imbalance

The comparison uses the prospective batch in Section 6.4. Its four cells have $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, and matching or product crosses, with 4096 shared replicate labels per cell. For each saved endpoint at $n_* = q^2$, compare

$$x = Z_c(n_*)/q, \qquad b = |\Delta_{n_*}|/(2q), \qquad p = S_0(\Delta_{n_*}/q).$$

Across all four cells, mean absolute error falls from 0.3961–0.4355 to 0.0174–0.0180, a 95.6–95.9% reduction. The configuration audit and treatment of shared labels are in Supplement sections SM7 and SM5.

6.3 Finite-scale errors

The original batch uses $q = 64, 128, 256, 512$, $\lambda = 0, 1, 2$, and matching or product crosses. It has respectively 128, 128, 256, and 1024 labels per cell. Feedback error compares the actual conditional cross response with $g(h_n, \Delta_n/q)$ at 16 prescribed phases. Overflow error is the maximum of $|Z_c/q - S_h(\Delta/q)|$ over the five prescribed offsets $h = -1, -1/2, 0, 1/2, 1$. Fig. 6 summarizes the nonzero-coupling cells.

At $q = 512$, all primary comparisons meet the prespecified error margins. The overflow-error failure probabilities have upper bounds 0.058–0.073. Cross-kernel cumulative distribution function (CDF) distance bounds are at most 0.130602. Supplement section SM5.1 gives the statistical intervals and error allocation.

6.4 Terminal diffusion and unconditional overflow

The prospective batch has $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, and matching or product crosses. Its four cells each have 4096 labels. The labels are shared across cells, giving 16384 endpoints. Each label is the sampling unit. The targets are the unconditional mean difference $\mathbb{E}[Z_c(q^2)/q] - \mathbb{E}[S_0(D_1)]$ and the Kolmogorov distance between $\mathcal{L}(\Delta_{q^2}/q)$ and $\mathcal{L}(D_1)$. The reference uses 32768 preserved stochastic differential equation (SDE) paths per λ, with 16384 Euler steps per path. The overflow mean is computed from this SDE reference.

All eight comparisons meet the prespecified error margins (Supplement section SM5).

6.5 Sensitivity to ties and internal candidate graphs

To test persistence and amplification of internal overflow, we vary the tie rule and internal candidate graph. We test $q = 128, 256, 512$, $\kappa = \lambda = 1$, product crosses, and 1024 trajectories per cell. Left ties always favor the left endpoint; fair ties use an independent coin. The four internal graphs are complete bipartite, the union of 16 fixed random permutations, a degree-two cycle, and a disconnected matching. All have uniform endpoint marginals. For the permutation graph, each cell averages four fixed graph pairs, with 256 trajectories per pair. The 24 cells contain 24576 endpoints. Supplement section SM9 gives the protocol, graph construction, shared labels, and all cell summaries.

The complete-bipartite left-tie mean 0.3958 estimates $\mathbb{E}[G_c/q]$. Corollary 2.4 gives its limiting target as

$$\mathbb{E}\big[S_0(D_\kappa) - |D_\kappa|/2\big].$$

The expectation averages the omitted overflow over random terminal imbalance. The value $S_0(0) \approx 0.6313$ in Fig. 1 instead evaluates the profile at zero imbalance, where community excess vanishes. Section 6.2 compares the profile with individual observed imbalances; Section 6.4 tests the unconditional total-overflow target $\mathbb{E}S_0(D_\kappa)$. Window averages in Supplement Table SM6 condition on $|\Delta|/q \le 1/8$ and retain variation within that window.

Mean omitted overflow is positive in all 24 tested configurations. Cycles and disconnected matchings amplify it (Table 2 and Fig. 5).

An additional analysis of the saved outcomes gives simultaneous 95% intervals for the 24 cell means and 36 graph contrasts within each size and tie rule. At $q = 512$ with left ties, the cycle-minus-complete difference is 1.9104, with interval $[1.6787, 2.1421]$. The matching-minus-complete difference is 8.0681, with interval $[7.8364, 8.2997]$. Supplement section SM9.4 gives the bounded-difference construction and both tie rules.

For complete sampling, the paired fair-minus-left mean difference in G_c/q is 0.02441, with Monte Carlo standard error 0.00072 at $q = 512$. Across the three sizes, the left-tie complete-bipartite means are $0.3864, 0.3989, 0.3958$; the corresponding fair-tie means

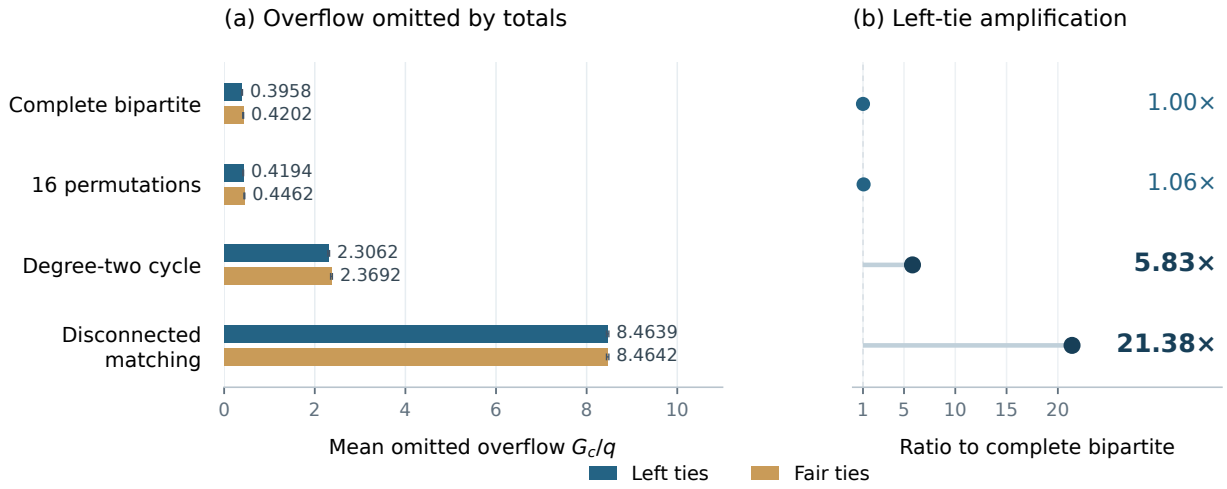


Fig. 5 Cycles and disconnected matchings amplify overflow omitted by community totals. Results use $q = 512$, $\kappa = \lambda = 1$, and 1024 trajectories per cell. Left: mean G_c/q for left and fair ties, with twice Monte Carlo standard error conditional on the fixed graphs. Right: left-tie sample means relative to complete-bipartite sampling; the dashed line marks ratio one. Cycle and matching means are 5.83 and 21.38 times the baseline. Table 2 gives the same means; Supplement Fig. SM1 shows all three tested sizes.

Table 1 Prospective comparisons at $q = 512$, $\kappa = 1$. Mean intervals bound microscopic mean overflow minus the SDE mean readout. CDF upper bounds concern terminal imbalance and have lower bound zero. All displayed limits are rounded outwards and include reference error.

λ	Cross kernel	Mean-difference interval	CDF upper bound
1	matching	$[-0.00840,\ 0.01010]$	0.06542
1	product	$[-0.00860,\ 0.00993]$	0.06500
2	matching	$[-0.00663,\ 0.00844]$	0.07021
2	product	$[-0.00671,\ 0.00834]$	0.07119

Table 2 Mean omitted overflow G_c/q at $q = 512$, $\kappa = \lambda = 1$, with 1024 trajectories per cell. Here $G_c = Z_c - |\Delta|/2$ at critical fill. Parentheses give Monte Carlo standard errors (MCSEs), conditional on the fixed graphs.

Internal graph	Left ties	Fair ties
Complete bipartite	0.3958 (0.00454)	0.4202 (0.00473)
16 permutations	0.4194 (0.00468)	0.4462 (0.00484)
Degree-two cycle	2.3062 (0.00993)	2.3692 (0.01001)
Matching	8.4639 (0.01320)	8.4642 (0.01320)

are $0.4105, 0.4226, 0.4202$. The normalized contribution increases across these sizes for both the cycle and the matching. With left ties, the cycle means rise from 1.4405 to 2.3062; the matching means rise from 3.9458 to 8.4639. Fair ties show the same increasing trend over the tested sizes. Uniform candidate marginals coexist with markedly different omitted overflow across the tested graphs.

7 Discussion

Internal overflow measures a mismatch between where load accumulates and where capacity remains available. The aggregation gap connects this mismatch to the within-community load distribution.

7.1 Finite-size approximation

The coupling estimate quantifies the comparison with product cross sampling. Quantitative profile attraction and tracking rates would connect the critical law to finite-size error estimates.

7.2 What aggregation removes

For any partition into communities and common capacity c, define excess load and unused capacity within community s by

$$E_s = \sum_{v \in C_s} (A_v - c)_+, \qquad U_s = \sum_{v \in C_s} (c - A_v)_+.$$

Since $M_s - |C_s|c = E_s - U_s$, elementary positive-part algebra gives the identity (1.1) stated in the introduction. Its absolute-deviation form is

$$G_c = \frac{1}{2} \sum_s \left\{ \sum_{v \in C_s} |A_v - c| - |M_s - |C_s|c| \right\}. \tag{7.1}$$

It is the gap from applying the convex positive-part function after aggregation.

At the critical count of our two-community model, $G_c = Z_c - |\Delta|/2$. Corollary 2.4 identifies its strictly positive limiting mean after division by q. The profile theorem therefore quantifies the local capacity mismatch that community totals cancel.

The topology comparisons suggest a mechanism based on the reach of a candidate pair. Complete-bipartite sampling can compare any left

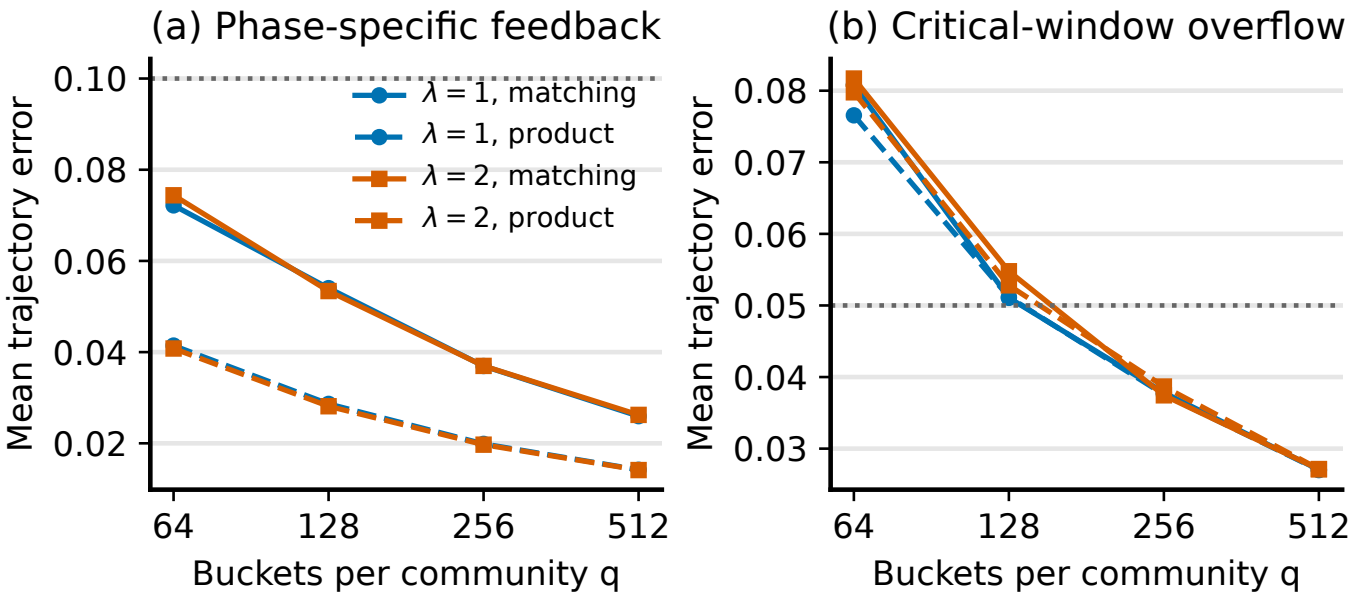


Fig. 6 Finite-scale errors from the original saved outcomes. Left: mean per-trajectory feedback root mean square (RMS) over 16 phases. Right: mean per-trajectory maximum overflow error over five offsets. Line styles distinguish matching and product kernels; markers distinguish $\lambda = 1, 2$. The horizontal lines mark the prespecified per-trajectory margins.

bucket with any right bucket in the same community. A degree-two cycle only compares neighboring buckets. A matching confines internal arrivals to fixed bucket pairs; cross arrivals provide occasional connections between communities. Restricted candidate access can leave excess load in one region while capacity remains unused elsewhere. Equation (1.1) records this coexistence through $\min\{E_s, U_s\}$.

Uniform endpoint marginals describe how often each bucket is offered. The internal graph determines which alternatives an arrival can compare. For fixed internal matchings, product cross sampling, and left or fair ties, Supplement section SM9.2 gives

$$\mathbb{E}G_c(n_*) = \{(2\pi)^{-1/2} + o(1)\}q^{3/2} \quad \text{when } c = q/2,\ \varepsilon_q q \to \lambda < \infty.$$

Replacing cross arrivals by internal arrivals changes expected omitted overflow by at most $2\varepsilon_q q^2 = O(q)$. The perturbation preserves the leading term under the experiment's weak cross sampling.

7.3 Implications for capacity monitoring and design

A community total records net excess $E_s - U_s$. It does not record the coexistence of excess demand and unused space. Capacity monitoring can retain the pair (E_s, U_s) or the bucket-load histogram to preserve information lost by the community total. For the no-departure model, these loads are cumulative assigned counts. In a capped implementation, occupancy counts must be accompanied by overflow counts to retain demand above capacity.

8 Conclusion

Under complete-bipartite internal sampling, community totals omit a positive order-q contribution to expected overflow at critical fill. The joint critical law combines a scalar imbalance diffusion with an attracting periodic load profile. The proof uses process coupling, profile contraction, and phase averaging. The aggregation identity identifies the missing capacity cost as excess load coexisting with unused space within a community. It gives a direct reason to retain internal load information in capacity monitoring.

Across the tested sizes and four internal graphs, mean omitted overflow is positive under both left and fair ties. Cycles and disconnected matchings amplify this contribution. Further questions concern overflow scales on sparse internal graphs, the joint critical law for fair ties, and adaptive routing with finite capacity.

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Competing interests

The authors declare that they have no competing interests or financial conflicts to disclose.

Data availability

The original data and code are publicly available at `https://github.com/19991111/critical-overflow-reproducibility`. The accompanying source archive includes the sensitivity experiments and supporting records (Supplement sections SM7 and SM9).

Declarations

OpenAI's Codex assisted with drafting, literature searches, proof checking, and preparation of code and figures. The authors take responsibility for all content.

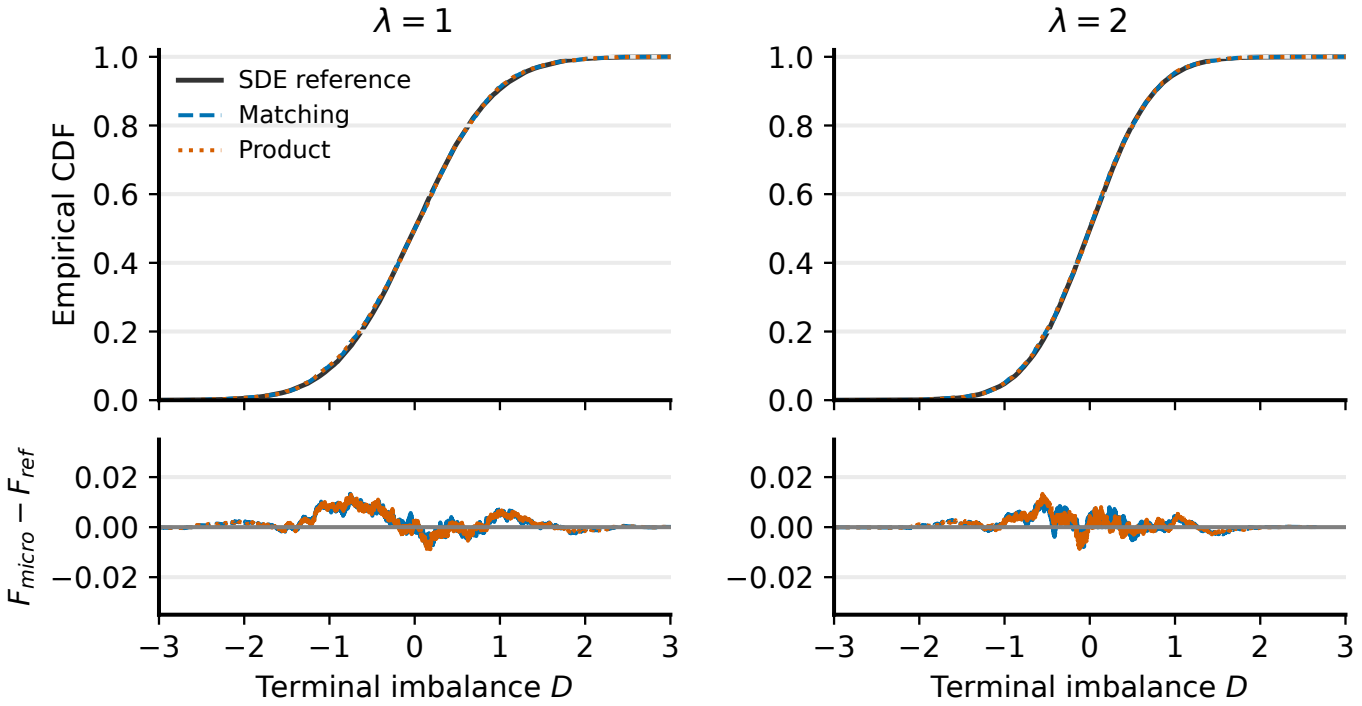


Fig. 7 Terminal imbalance: the two prospective microscopic empirical CDFs and the independently generated, preserved SDE reference for each λ. The lower panels show microscopic minus reference empirical CDFs at all jumps. Each microscopic cell has 4096 labels; each reference has 32768 paths. The numerical and sampling allowances enter the distance bounds in Table 1.

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